CONSTRUCTIBLE SUBSETS PULL BACK ALONG AN OPEN IMMERSION

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The criterion expressing constructibility through inverse images of retrocompact opens. Invoked at the start of the counted proof (source lines 2837-2838). Reduces the assertion to retrocompactness of the traces of retrocompact opens.

Source reference. Original lemma lemma-inverse-images-constructibles: sources/stacks-topology/pinned-original-source.tex, source lines 2808–2808. Navigation pointer only: the counted text refers to the Stacks item lemma-inverse-images-constructibles, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let $U \subset X$ be open. For a constructible set $E \subset X$ the intersection $E \cap U$ is constructible in U.*

Proof. Suppose that $V \subset X$ is retrocompact open in X. It suffices to show that $V \cap U$ is retrocompact in U by Lemma lemma-inverse-images-constructibles. To show this let $W \subset U$ be open and quasi-compact. Then W is open and quasi-compact in X. Hence $V \cap W = V \cap U \cap W$ is quasi-compact as V is retrocompact in X. □

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